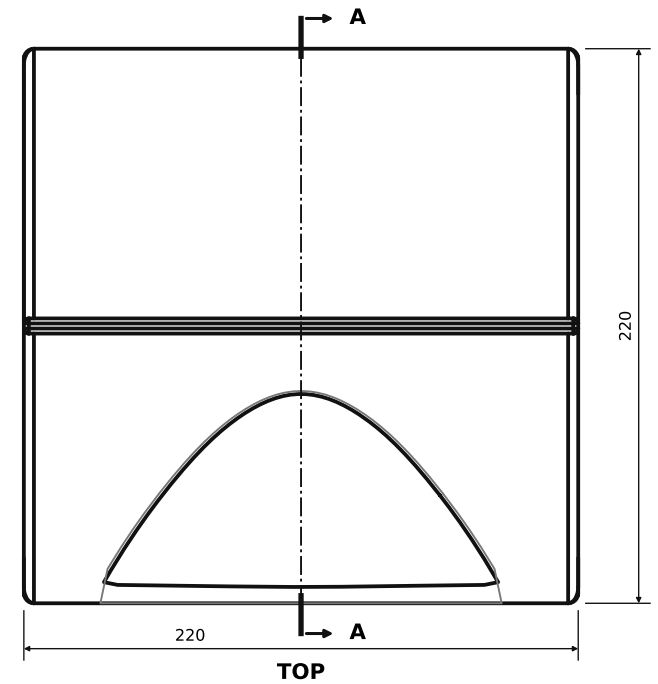
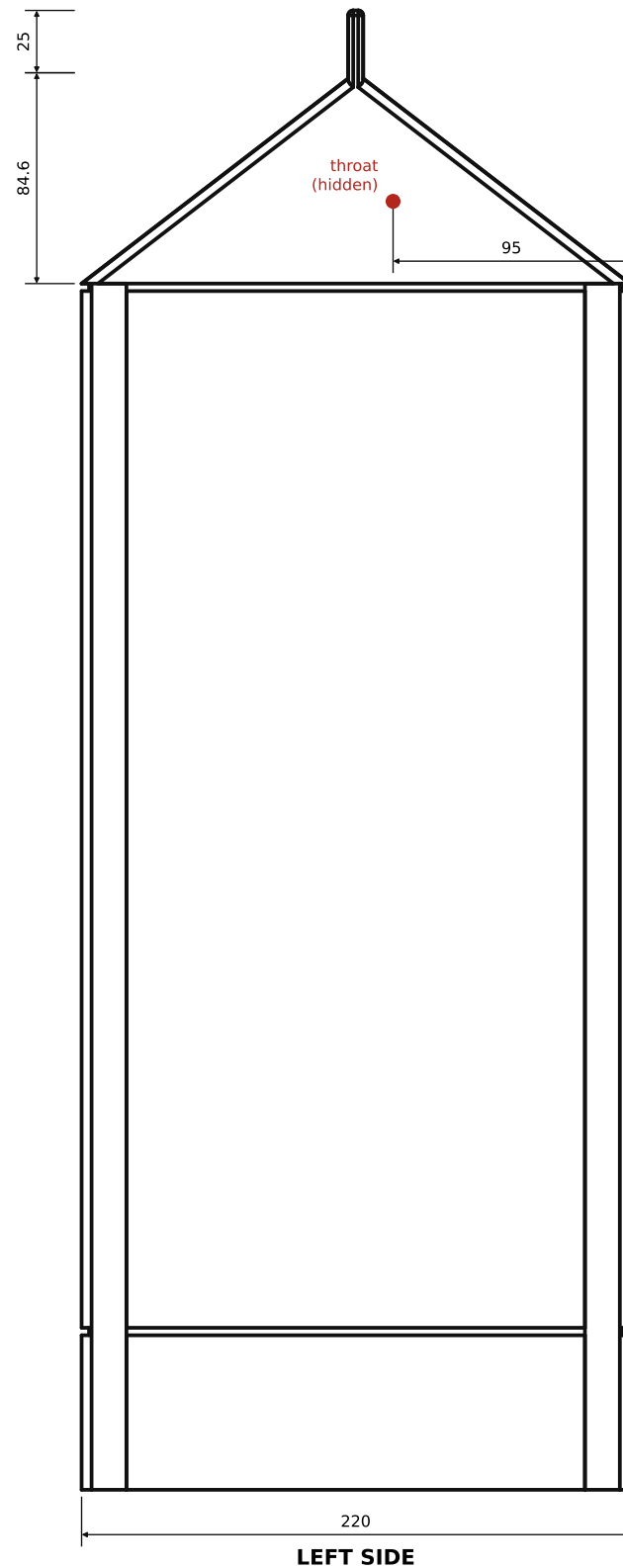
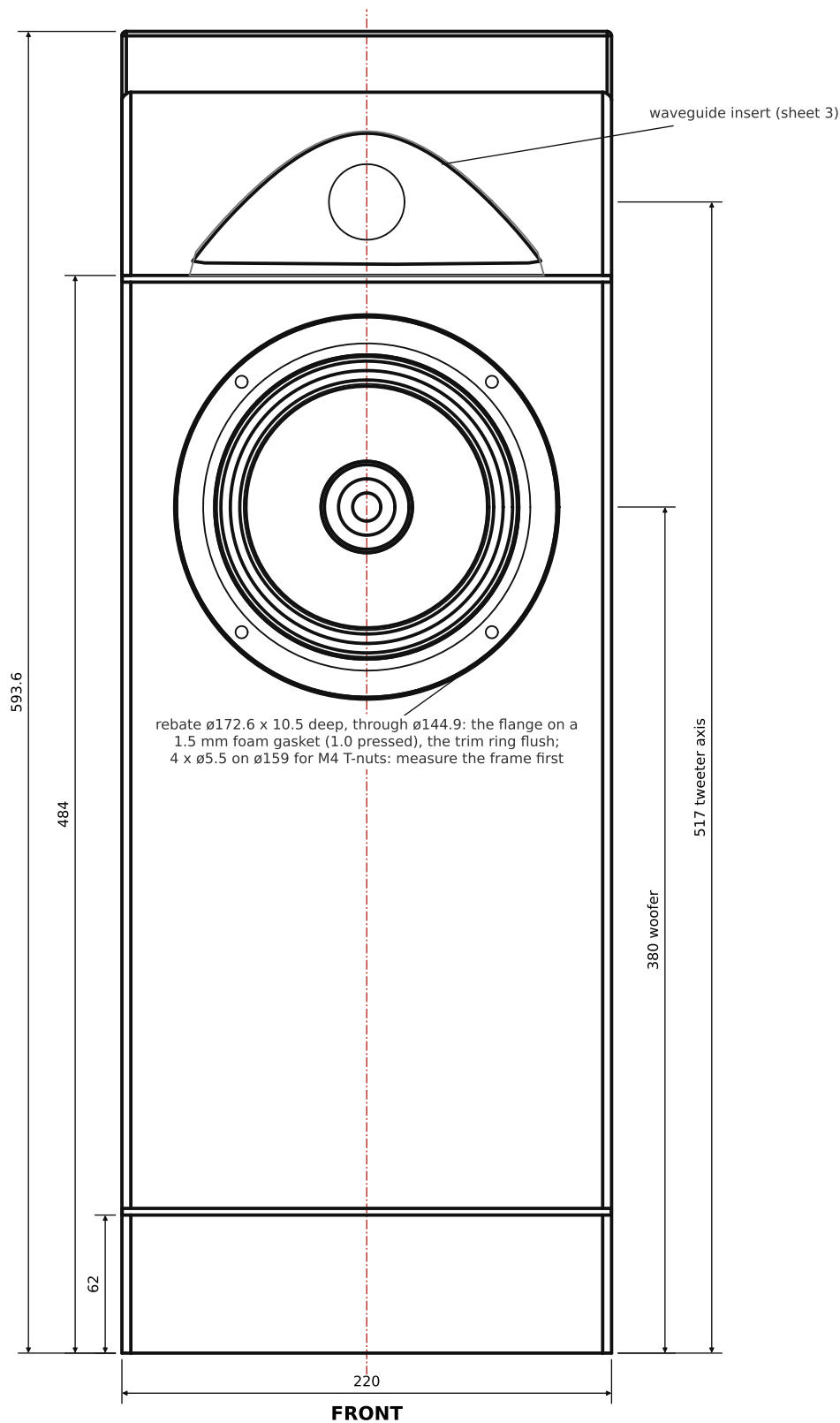




black: the waveguide's mouth, where its lip meets the roof
grey: the insert's seam

Notes

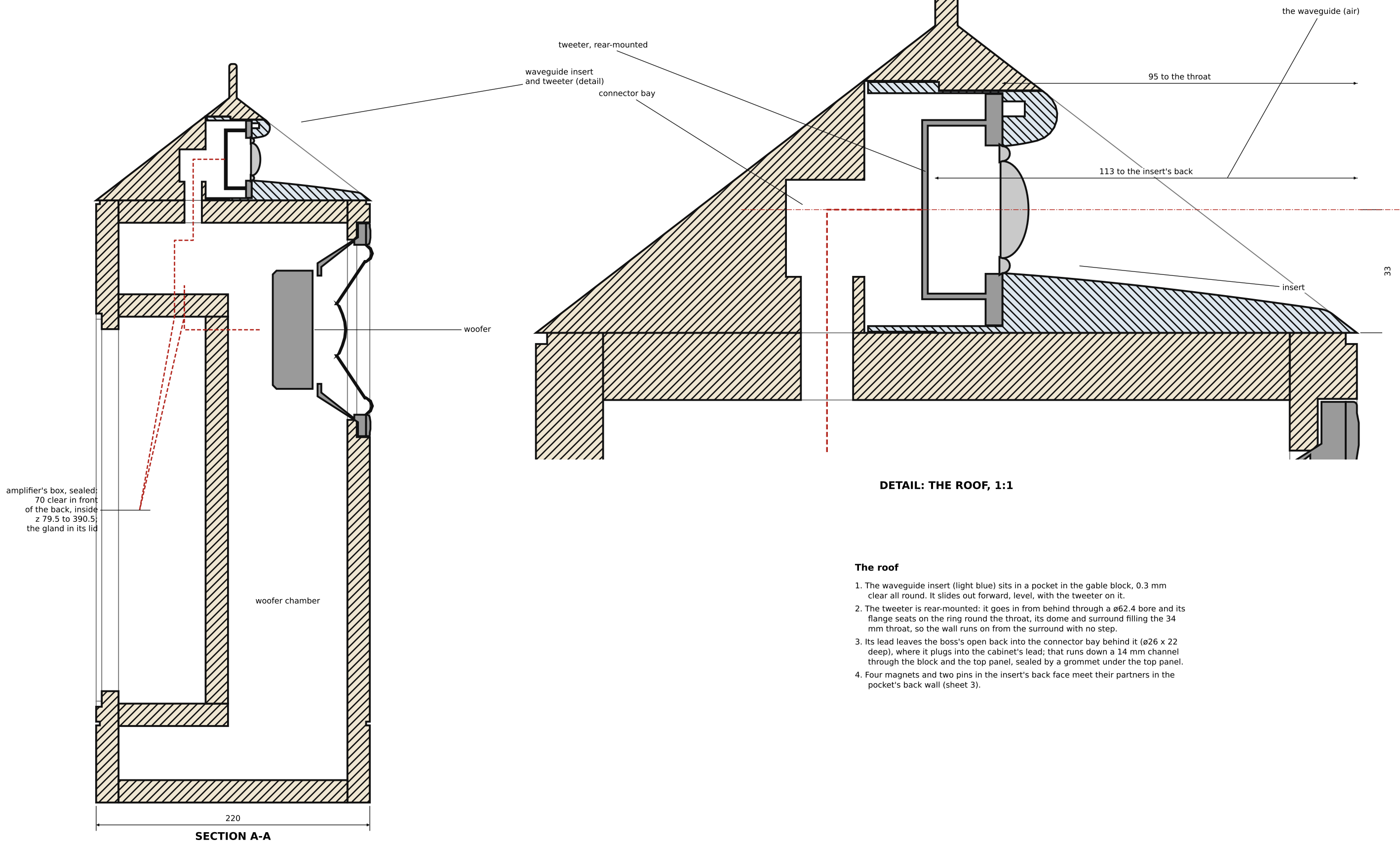
1. Cabinet: 18 mm Baltic birch, glued. The vertical corners and the gable's hips are rounded 4 mm, the fin's edges 2 mm. 3 x 3 mm shadow lines at z 62 and z 481.
2. The drivers sit flush: each frame on a 1.5 mm closed-cell foam gasket in its rebate, M4 button heads into T-nuts, and a 3 mm printed trim ring over the flange (stl/trim-ring-*.stl: a channel in its back clears the screw heads), held by three 5 mm dots of neutral-cure silicone, 0.8 reveal all round.
3. The tweeter is at the throat of a waveguide insert in the roof (sheet 3): axis z 517, 95 behind the front face. The shape is set by simulation (fab/out-bookshelf/acoustics/waveguide).
4. The back (the amplifier) is drawn on sheet 4 with the wiring; the Nutrition Facts are printed on the right side.
5. Finish: colour coat and 2K clear; the Facts printed between them.

earmilk bookshelf | General arrangement

Front, left side and top at 1:3. Heights from the floor; depths from the front face.

mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned

Tolerances: panels ± 0.5 , pocket depths ± 0.2 , holes $+0.2/-0$, prints ± 0.15 . Check bought parts first.



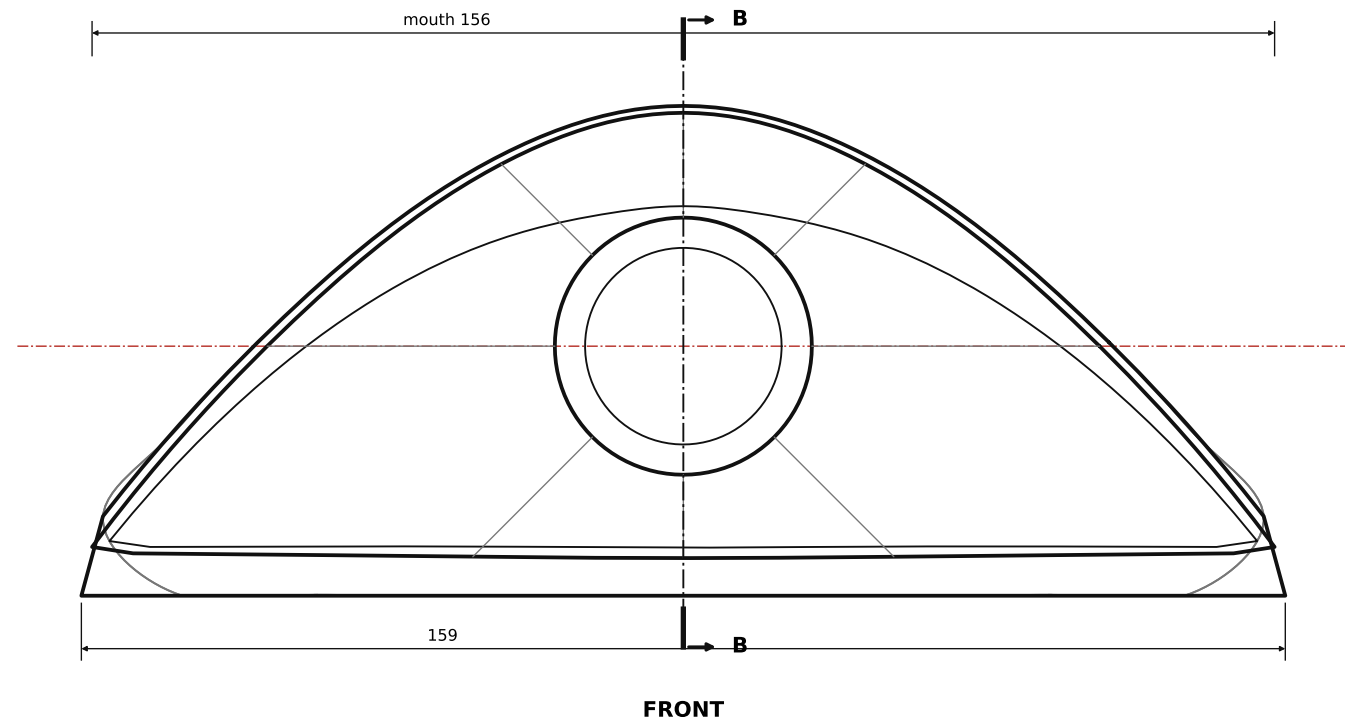
The roof

1. The waveguide insert (light blue) sits in a pocket in the gable block, 0.3 mm clear all round. It slides out forward, level, with the tweeter on it.
2. The tweeter is rear-mounted: it goes in from behind through a $\varnothing 62.4$ bore and its flange seats on the ring round the throat, its dome and surround filling the 34 mm throat, so the wall runs on from the surround with no step.
3. Its lead leaves the boss's open back into the connector bay behind it ($\varnothing 26 \times 22$ deep), where it plugs into the cabinet's lead; that runs down a 14 mm channel through the block and the top panel, sealed by a grommet under the top panel.
4. Four magnets and two pins in the insert's back face meet their partners in the pocket's back wall (sheet 3).

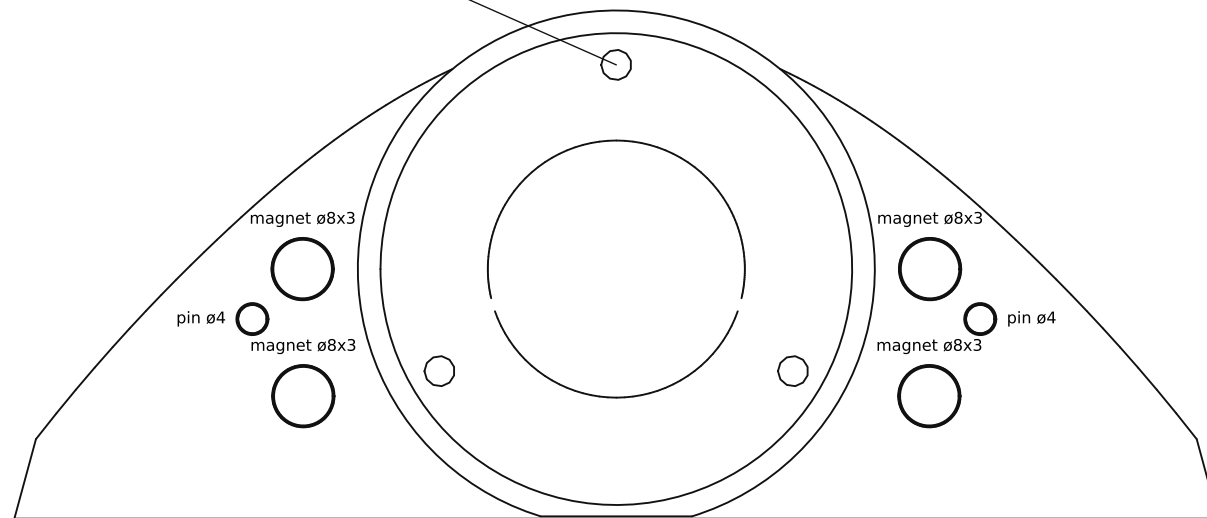
earmilk bookshelf | Section on the centreline

Section A-A on the centre plane (x 110), from the left, at 1:3; the roof at 1:1. The insert hatched the other way.

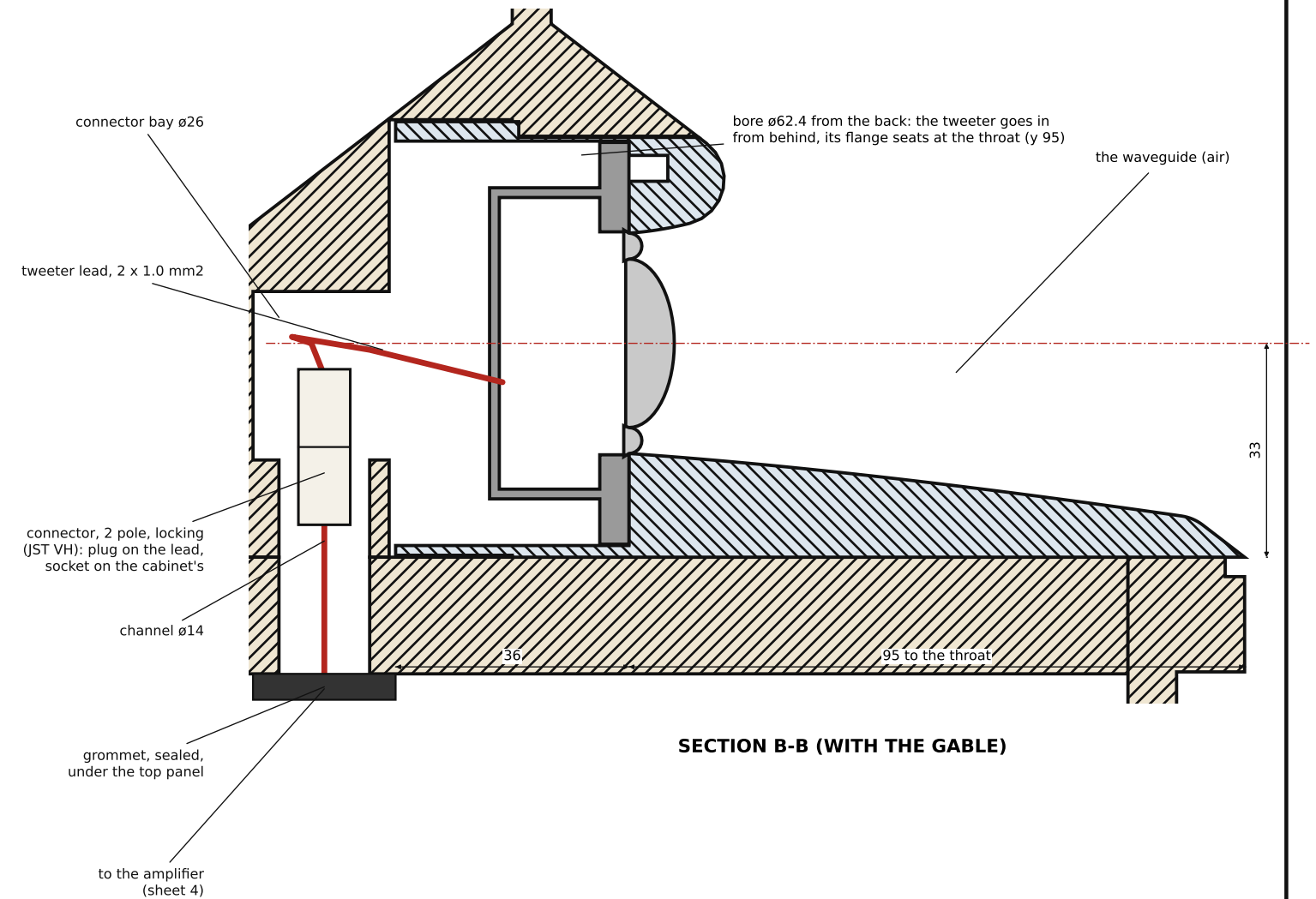
mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned
Tolerances: panels ± 0.5 , pocket depths ± 0.2 , holes $+0.2/-0$, prints ± 0.15 . Check bought parts first.



3 x $\varnothing 4 \times 6$ on $\varnothing 54$ in the seat, for
M3 x 5.7 heat-set brass inserts: measure the tweeter first



BACK (THE FACE THAT MEETS THE POCKET)



Fitting the tweeter and the insert

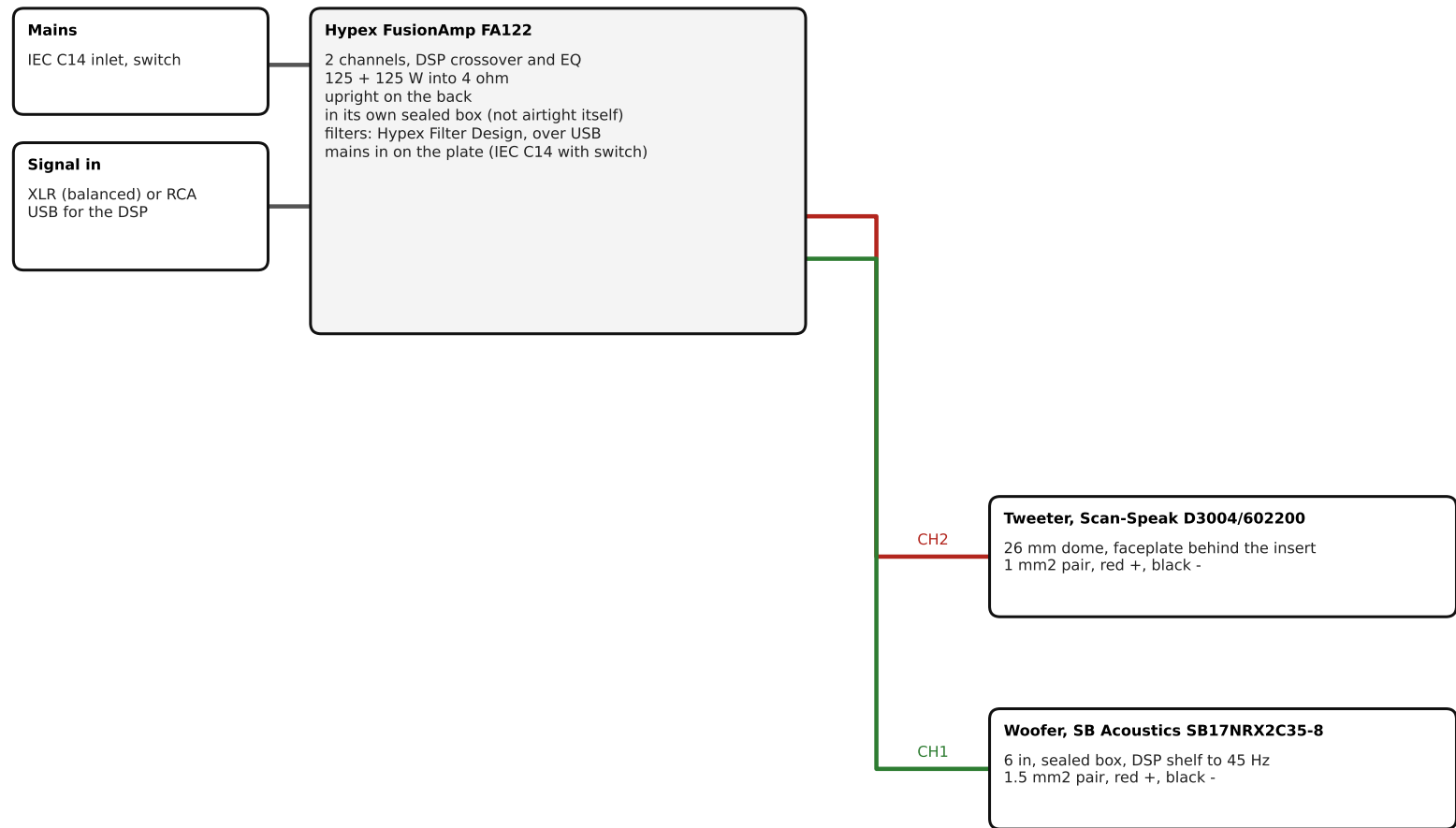
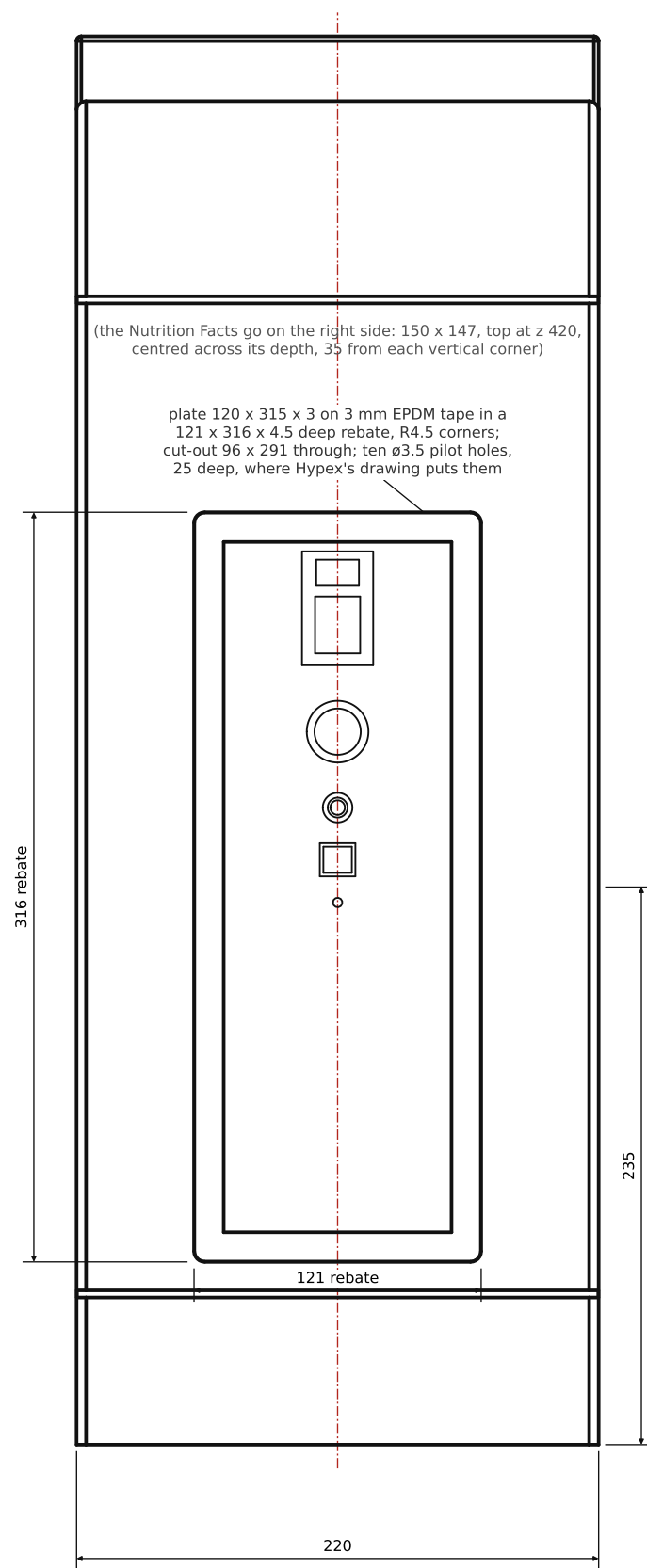
1. Solder the tweeter's lead (2 x 1.0 mm2, 120 mm) to its tabs and crimp the plug half of a two-pole locking connector (JST VH) on its end.
2. Melt 3 M3 x 5.7 heat-set brass inserts into the seat round the throat ($\varnothing 4 \times 6$ holes on $\varnothing 54$). Pass the tweeter in from behind through the $\varnothing 62.4$ bore, dome first, and seat its flange on a 0.5 mm foam gasket; screw it to the inserts.
3. Glue the magnets into the insert's back with epoxy, polarity marked, and their partners into the pocket's back wall, opposite poles out. Press the two pins into the insert.
4. The cabinet's lead ends in the socket 100 mm above the grommet. With the insert out, draw the socket forward out of the bay through the empty pocket, plug the tweeter's lead into it, and tuck both and the slack back into the bay.
5. Slide the insert in, level, until the pins seat and the magnets pull it home: its face flush with the roof, the seam even.
6. Service: lift it out with a suction lifter on its flat border (or a pull loop under its lower edge), unplug, and the tweeter comes out with it.
7. Fixings, from the centreline and the insert's base (mirrored in the pocket's wall): magnets ± 41.4 across, 16.2 up; ± 41.5 across, 33.0 up; pins ± 48.1 across, 26.4 up. Magnets $\varnothing 8 \times 3$ in $\varnothing 8.2 \times 3.3$ holes, epoxied, poles marked; pins $\varnothing 4 \times 8$ pressed into $\varnothing 3.9 \times 4.5$ in the insert, a slip fit in $\varnothing 4.2 \times 4.5$ in the wall.

earmilk bookshelf | Waveguide insert and tweeter mount

The insert at 1:1: front, section on the axis, back; the pocket; fitting and service.

mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned

Tolerances: panels ± 0.5 , pocket depths ± 0.2 , holes $+0.2/-0$, prints ± 0.15 . Check bought parts first.

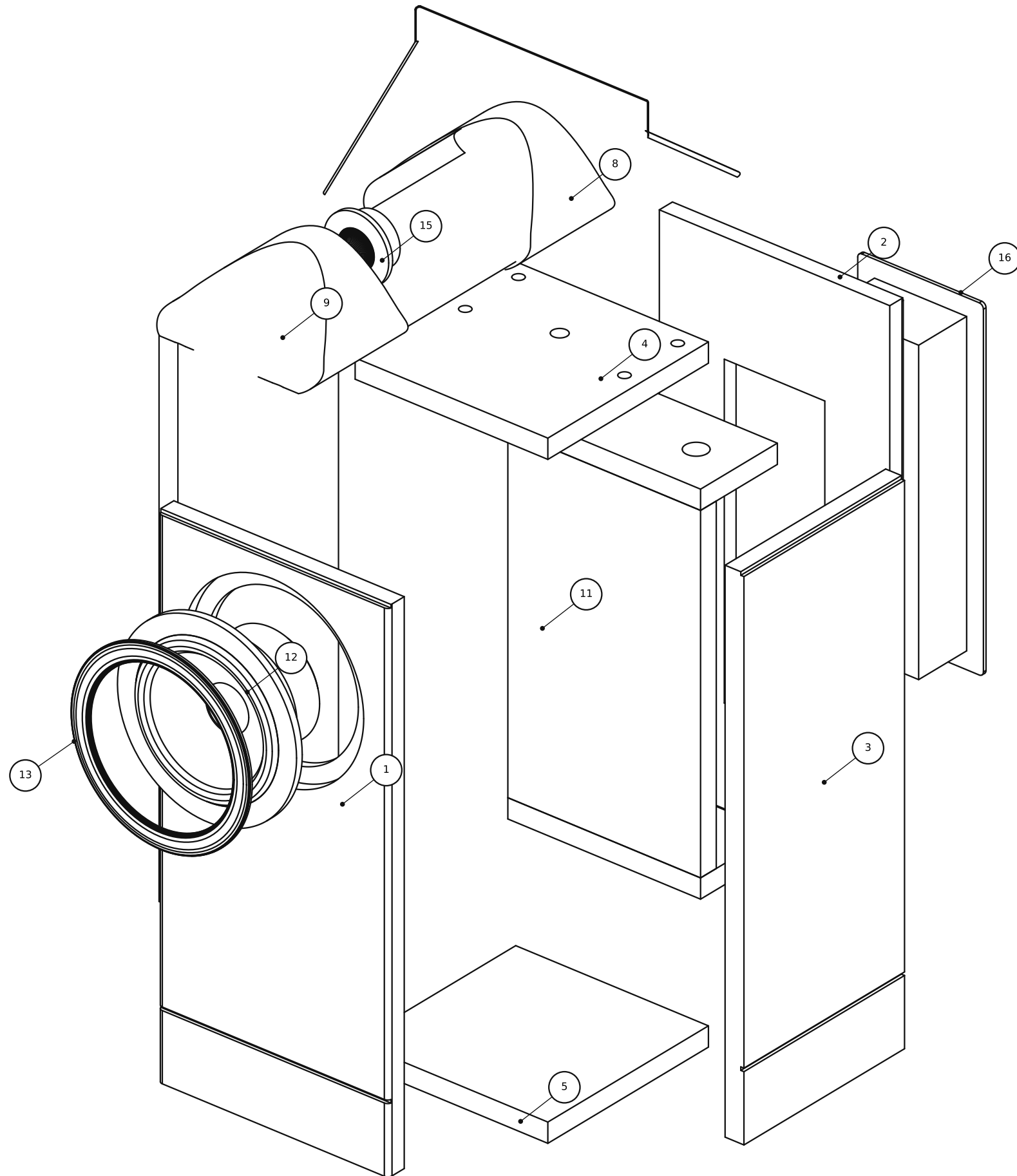


Runs, connectors and seals

1. Woofer (CH1): from the amplifier box's gland to the woofer's terminals, 0.4 m of 1.5 mm2.
2. Tweeter (CH2): from the gland up behind the woofer's magnet to the top panel, through the 14 mm channel in the top panel and the gable block, 0.6 m of 1.0 mm2, ending in the socket of a two-pole locking connector (JST VH) in the bay behind the insert.
3. Seal the tweeter cable where it passes the top panel (grommet and sealant): the box is sealed, and the insert's pocket is open to the room through its seam.
4. The amplifier box's lid: an M20 x 1.5 long-thread (15 mm) gland with a two-hole seal for the two pairs, in a ø20.5 hole 144 from the left side's inner face and 43 from the lid's front edge, its lock nut in a ø32 x 8 counterbore under the lid; seal it.
5. The crossover (2.4 kHz, LR4), the woofer's shelf to 45 Hz (+8.3 dB) and the delays are set in the DSP. Wire red to + throughout.

earmilk bookshelf | Back and wiring

The back at 1:3 and the amplifier, drivers, cables and connectors. Active: the amplifier's DSP is the crossover.



Parts

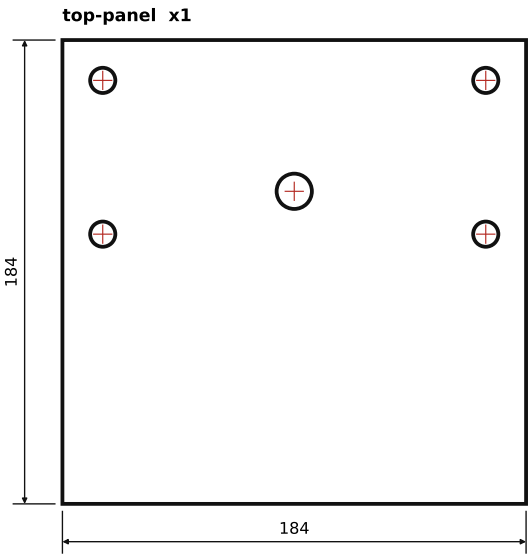
- ① **Front baffle x1**
18 mm Baltic birch, CNC: the woofer's cutout and rebate
- ② **Back x1**
18 mm birch: the amplifier's cutout and rebate
- ③ **Sides x2**
18 mm birch; the Facts printed on the right one
- ④ **Top panel x1**
18 mm birch, the tweeter cable's 14 mm hole, four dowel holes
- ⑤ **Bottom panel x1**
18 mm birch
- ⑧ **Gable block x1**
laminated birch or printed whole: the insert's pocket and the cable channel
- ⑨ **Waveguide insert x1**
printed (SLA, MJF) or CNC from solid; magnets, pins
- ⑪ **Amplifier box x3**
18 mm birch floor, lid (gland) and front, glued and sealed
- ⑫ **Woofer x1**
SB Acoustics SB17NRX2C35-8, 6 in coated paper
- ⑬ **Trim ring x1**
printed, satin black or gunmetal, over the frame and its screws
- ⑮ **Tweeter x1**
Scan-Speak Illuminator D3004/602200, faceplate behind the insert
- ⑯ **Amplifier x1**
Hypex FusionAmp FA122, DSP, 2 channels

earmilk bookshelf | Exploded view and parts

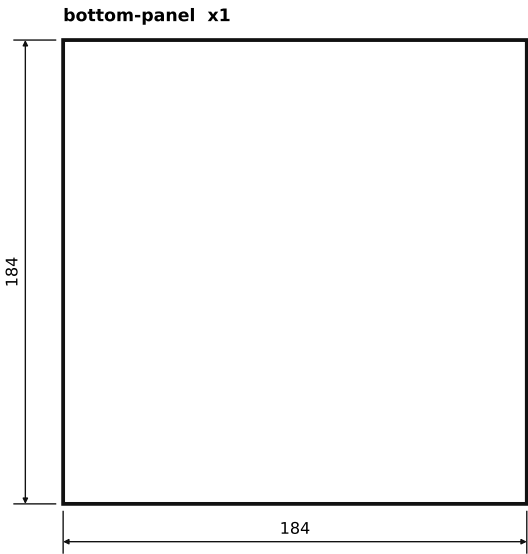
Isometric, not to scale: the parts pulled apart along the way they go together, and what each one is.

mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned

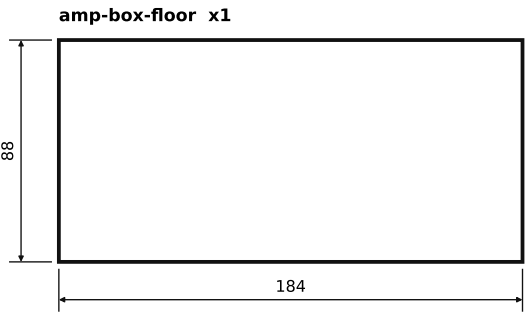
Tolerances: panels ± 0.5 , pocket depths ± 0.2 , holes $+0.2/-0$, prints ± 0.15 . Check bought parts first.



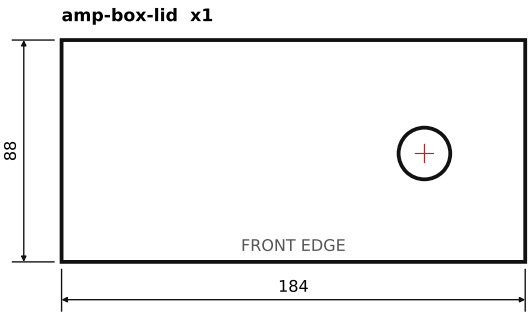
ø14 at (92, 124): cut inside
ø10 at (16, 107): drill ø10, 10 deep
ø10 at (168, 107): drill ø10, 10 deep
ø10 at (16, 168): drill ø10, 10 deep
ø10 at (168, 168): drill ø10, 10 deep
upper face up (front edge at the bottom of the drawing); dowels
register the gable block; the tweeter cable's hole gets a grommet



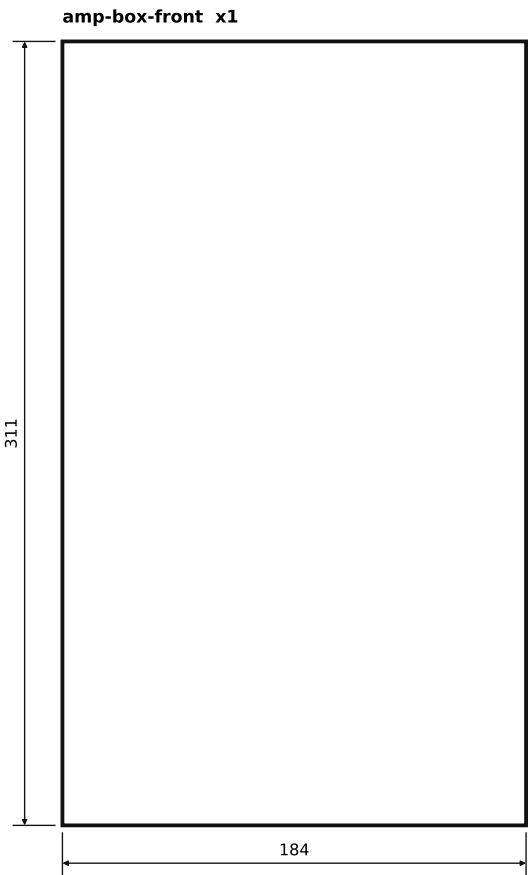
either face



the amplifier box's floor, z 61.5 to 79.5, against the back



ø20.5 at (144, 43): cut inside
the amplifier box's lid, z 390.5 to 408.5, its front edge (the
bottom of the drawing) over the box's front; the gland's ø20.5
hole 144 from the left, 43 from the front edge; sealed after
wiring



the amplifier box's front, between floor and lid, 70 in front of
the back's inner face; glue and seal all round

earmilk bookshelf | Panel details

Each panel seen from its outer (upper) face, from its lower-left corner as its DXF; holes from that corner. 18 mm Baltic birch.

mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned
Tolerances: panels ±0.5, pocket depths ±0.2, holes +0.2/-0, prints ±0.15. Check bought parts first.